

U.G.C. NET Exam

COMPUTER

(Solved with Explanation)

1. In 8085 microprocessor which of the following flag(s) is (are) affected by an arithmetic operation?

(a) AC flag only (b) CY flag only
(c) Z flag only (d) AC, CY, Z flags

Ans : (d) AC is auxiliary – carry flag, cy is carry flag & z is zero flag. All these flags will be affected during arithmetic operation.

2. In 8085 microprocessor the address bus is of bits

(a) 4 (b) 8 (c) 16 (d) 32

Ans : (c) In 8085 microprocessor 16 bits are used for address bus & 65,536 ($2^{16} = 65,536$) different memory location are possible.

3. In the architecture of 8085 microprocessor match the following :

A.	Processing unit	i.	Interrupt
B.	Instruction unit	ii.	General purpose Register
C.	Storage and Interface unit	iii.	ALU
D.		iv.	Timing control and

Code :

	A	B	C
(a)	iv	i	ii
(b)	iii	iv	ii
(c)	ii	iii	i
(d)	i	ii	iv

Ans : (b)

- ALU is the arithmetic logic unit and it involves processing of input desired output.
- Timing and control instruction are covered in instruction unit of microprocessor.
- There are some general purpose register in storage & interface unit
- While an interrupt is signal to the processor which required attention from processor, an interrupt is serviced on the basis of priority & need

4. Which of the following addressing mode is best suited to access elements of an array of contiguous memory locations?

(a) Indexed addressing mode
(b) Base Register addressing mode
(c) Relative address mode
(d) Displacement mode

n. November, 2017

R SCIENCE

anation Paper–III)

Ans : (a) Indexed addressing mode is best suited for accessing an array in contiguous memory location.

5. Which of the following is correct statement?

- (a) In memory-mapped I/O, the CPU can manipulate I/O data residing in interface registers that are not used to manipulate memory words
- (b) The isolated I/O method isolates memory and I/O addresses so that memory address range is not affected by interface address assignment
- (c) In asynchronous serial transmission of data the two units share a common clock
- (d) In synchronous serial transmission of data two units have different clocks

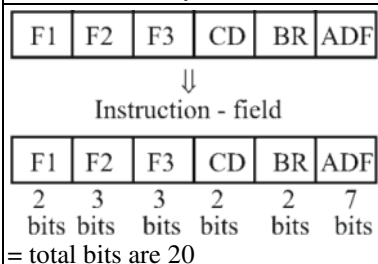
Ans : (b) The Isolated I/O method isolates memory & I/O addresses so that memory address range is not affected by interface address assignment.

- Memory based I/O uses same address space for memory & I/O devices
- In asynchronous serial transfer of data the two units do not share a common clock.
- In synchronous serial transfer of data the two units share a common clock.

6. A micro-instruction format has micro-ops field which is divided into three subfields F₁, F₂, F₃ each having seven distinct micro-operations, condition field CD for four status bits, branch field BR having four options used in conjunction with address field ADF. The address space is of 128 memory locations. The size of micro-instruction is :

- (a) 17 bits
- (b) 20 bits
- (c) 24 bits
- (d) 32 bits

Ans : (b) Microprocessor instruction format, which is divided into three subfields F₁, F₂, F₃ each having seven distinct micro-operation condition field CD for four status bits, branch fields BR having four options used in conjunction with address field ADF. The address space is of 128 memory locations i.e.



7. Consider the following four schedules due to three transactions (indicated by the subscript) using read and write on a data item X, denoted by $r(X)$ and $w(X)$ respectively. Which one of them is conflict serializable?

$S_1 : r_1(X); r_2(X); w_1(X); r_3(X); w_2(X)$

$S_2 : r_2(X); r_1(X); w_2(X); r_3(X); w_1(X)$

$S_3 : r_3(X); r_2(X); r_1(X); w_2(X); w_1(X)$

$S_4 : r_2(X); w_2(X); r_3(X); r_1(X); w_1(X)$

(a) S_1

(b) S_2

(c) S_3

(d) S_4

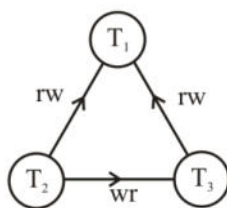
Ans : (d)

⇒ A Schedule is called conflict serializable if it can be transformed into a serial schedule by swapping non-conflicting operations.

For a schedule to be conflict serializable, there should be no cycle present in its precedence graph.

Option- (d) Conflict serializable

T_1	T_2	T_3
	$r(x)$	
	$w(x)$	
		$r(x)$
$r(x)$		
$w(x)$		



No Cycle

Option (d) is a conflict serializable schedule.

8. Suppose a database schedule S involves transactions $T_1, T_2, \dots, T_n$. Consider the precedence graph of S with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?

(a) Topological order

(b) Depth - first order

(c) Breadth - first order

(d) Ascending order of transaction indices

Ans : (a) For a schedule, we check its serializability by drawing a precedence graph & find its topological order, precedence graph of schedule must not contain any cycle to be conflict free.

9. If every non-key attribute is functionally dependent on the primary key, then the relation is in

(a) First normal form

(b) Second normal form

(c) Third normal form

(d) Fourth normal form

Ans : (b) 2NF is a relation R is in second normal form (2NF) if and only if it is in 1NF and every non key attribute is fully dependent on primary key ex : relation R(xyz) with functional dependencies ($x \rightarrow y$, $y \rightarrow z$, $x \rightarrow z$). The relation is in 2NF but not in 3NF because of every non key attribute is transitively dependent on the primary key. Here {X} will be candidate key.

10. Consider a relation R (A, B, C, D, E, F, G, H), where each attribute is atomic, and following functional dependencies exist

$CH \rightarrow G$

$A \rightarrow BC$

$B \rightarrow CFH$

$E \rightarrow A$

$F \rightarrow EG$

The relation R is

- (a) in 1NF but not in 2NF
- (b) in 2NF but not in 3NF
- (c) in 3NF but not in BCNF
- (d) in BCNF

Ans : (a) $A^+ \rightarrow$ All attribute except (d) similarly for other keys we can find closure, but (d) can't be derived from any key & it must be added to all keys to be derived from. That's why relation is in 1NF, Since there is no partial dependency, So this relation is not in 2NF.

11. Given two relations $R_1(A, B)$ and $R_2(C, D)$, the result of following query

Select distinct A, B

From R_1, R_2

is guaranteed to be same as R_1 provided one of the following condition is satisfied.

- (a) R_1 has no duplicates and R_2 is empty
- (b) R_1 has no duplicates and R_2 is non - empty
- (c) Both R_1 and R_2 have no duplicates
- (d) R_2 has no duplicates and R_1 is non - empty

Ans : (b) In this query first we will take Cartesian product R_1, R_2 for this $0-R_2$ then select distinct A, B from Cartesian product of R_1, R_2 and select A, B for R_2 for this query $A, B \rightarrow$ No duplicate A, B by combination condition for both we will get the right condition.

12. Consider a schema R(A, B, C, D) and following functional dependencies.

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$

$D \rightarrow B$

Then decomposition of R into $R_1(A, B)$, $R_2(B, C)$ and $R_3(B, D)$ is

- (a) Dependency preserving and lossless join
- (b) Lossless join but not dependency preserving
- (c) Dependency preserving but not lossless join
- (d) Not dependency preserving and not lossless join

Ans : (a) Schema $R(A, B, C, D)$ is decomposed into tree relation $\rightarrow R_1(A, B), R_2(B, C) \& R_3(B, D)$

Now dependencies derived from $R_1(A, B)$ are $A \rightarrow B, B \rightarrow C$ but c is not attribute here is this relation, similarly do for $R_2 \& R_3$ & in the last all the dependencies are preserved & it is a lossless decomposition.

13. Which of the following is not a component of memory tube display?

- (a) Flooding gun
- (b) Collector
- (c) Ground
- (d) Liquid Crystal

Ans : (d) Components of memory tube display are (1) Floor gun, (2) primary gun (3) writing beam (4) collector (5) storage grid (6) screen (7) flood electron (8) ground or base (9) Focusing & deflection system.
 $\Rightarrow$ Only liquid crystal is not enlisted here and it is not the display component in memory tube.

14. Which of the following is not true in case Oblique Projections?

- (a) Parallel projection rays are not perpendicular to the viewing plane
- (b) Parallel lines in space appear parallel on the final projected image
- (c) Used exclusively for pictorial purposes rather than formal working drawings
- (d) Projectors are always perpendicular to the plane of projection

Ans : (d) Oblique projection is a simple type of technical drawing of graphical projection used for producing two-dimensional images of three dimensional objects.

Some of the **properties of oblique projections:**

1. Parallel projections rays are not perpendicular to viewing plane, but strike the projection plane at an angle other than 90° .
2. Parallel lines in space appear parallel on the projected image (Same for orthographic projection).
3. Because of its simplicity, oblique projection is used exclusively for pictorial purpose rather than for formal working drawing.

15. With respect to CRT, the horizontal retrace is defined as :

- (a) The path an electron beam takes when returning to the left side of the CRT
- (b) The path an electron beam takes when returning to the right side of the CRT
- (c) The technique to turning the electron beam off while retracing
- (d) The technique of turning the electron beam on/off while retracing

Ans : (a) In computer graphics horizontal retrace is defined as the path an electron beam takes when returning to the left side of CRT.

16. Find the equation of the circle $x^2 + y^2 = 1$ in terms of $x'y'$ coordinates, assuming that the xy coordinate system results from a scaling of 3 units in the x' direction and 4 units in the y' direction.

- (a) $3(x')^2 + 4(y')^2 = 1$
 (b) $\left(\frac{x'}{3}\right)^2 + \left(\frac{y'}{4}\right)^2 = 1$
 (c) $(3x')^2 + (4y')^2 = 1$
 (d) $\frac{1}{3}(x')^2 + \frac{1}{4}(y')^2 = 1$

Ans : (b) Let $f[x, y] = 0$ be the equation for a curve in rectangular coordinates.

To scale the curve by s , the new equation would be

$$f[x/s, y/s] = 0$$

Let $f[\theta, r] = 0$ be the equation for a curve in polar coordinate. To scale it by s , the new formula is $f[r/s, \theta] = 0$

Replace x to $x'/3$ and y to $y'/4$, and substitute the value in the $x^2 + y^2 = 1$.

17. Find the normalization transformation that maps a window whose lower left corner is at (1 1) and upper right corner is at (3, 5) onto a view port that is the entire normalized device screen

$$\begin{array}{ll} \text{(a)} \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & \frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 1 \end{pmatrix} & \text{(b)} \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & -\frac{1}{4} & \frac{1}{4} \\ 1 & 1 & 1 \end{pmatrix} \\ \text{(c)} \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & \frac{1}{4} & \frac{1}{4} \\ 1 & 0 & 0 \end{pmatrix} & \text{(d)} \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{4} & -\frac{1}{4} \\ 1 & 0 & 0 \end{pmatrix} \end{array}$$

Ans : (a) The window parameters are $w_{x \min} = 1, w_{y \min} = 1, w_{x \max} = 3, w_{y \max} = 5$

view port parameters are $v_{x \min} = 0, v_{y \min} = 0, v_{x \max} = 1, v_{y \max} = 1$

hence $s_x = 1/2, s_y = 1/4$

So N using =

$$N = \begin{pmatrix} 1 & 0 & v_{x \min} \\ 0 & 1 & v_{y \min} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -w_{x \min} \\ 0 & 1 & -w_{y \min} \\ 0 & 0 & 1 \end{pmatrix}$$

$$N = \begin{pmatrix} 1/2 & 0 & -1/2 \\ 0 & 1/4 & -1/4 \\ 0 & 0 & 1 \end{pmatrix}$$

18. The three aspects of quantization, programmers generally concerned with are :

- (a) Coding error, sampling rate and Amplification
 (b) Sampling rate, Coding error and Conditioning
 (c) Sampling rate, Aperture time and Coding error
 (d) Aperture time, Coding error and Strobing

Ans:(c) The three aspects of quantization, programmers generally concerned with are-

- (1) Sampling rate = The number of samples per second.
- (2) Aperture time = Aperture time is the sampling rate defined as the number of the second per sample.
- (3) Coding error = The difference b/w the conditional signal and the quantized signal.

So option (c) is correct.

19. The logic of pumping lemma is an example of.....

- (a) iteration
- (b) recursion
- (c) the divide and conquer principle
- (d) the pigeon - hole principle

Ans : (d) Pigeon - hole principles states that if there n pigeons fly into m hole and $n > m$ then at least one hole must contain more than one pigeons. And logic of pumping lemma states that finite state automation can assume only a finite number of states & because there are infinitely many input sequence, by the pigeon hole principle there must be at least one state to which the automata returns over and over again.

20. Heap allocation is required for languages that :

- (a) use dynamic scope rules
- (b) support dynamic data structures
- (c) support recursion
- (d) support recursion and dynamic data structures

Ans : (b) Heap allocation is required for language which support dynamic data structure.

21. Pumping lemma for regular language is generally used for proving :

- (a) whether two given regular expressions are equivalent
- (b) a given grammar is ambiguous
- (c) a given grammar is regular
- (d) a given grammar is not regular

Ans : (d) Pumping lemma is used to prove a language is regular or not means pumping lemma is used as proof for irregularity of a language. Thus, if a language is regular it always satisfies pumping lemma. If there exists at least one string made from pumping which is not in L , then L is surely not regular.

22. Which of the following problem is undecidable?

- (a) To determine if two finite automata are equivalent
- (b) Membership problem for context free equivalent
- (c) Finiteness problem for finite automata
- (d) Ambiguity problem for context free grammar

Ans : (d) Ambiguity problem for context free grammar is undecidable. The problem is undecidable if there is no turing machine will always halt in finite amount of time.

23. Finite state machine can recognize language generated by

- (a) Only context free grammar
- (b) Only context sensitive grammar
- (c) Only regular grammar
- (d) any unambiguous grammar

Ans : (c) Finite state machine can recognize language generated by regular grammar only.....would wrong.....as some CFG (type -2), CSG (type-1), type -0 grammars also generate regular language. Hence the "only" in option (c) makes. So answer option (c)

24. The language $L = \{b^i b c^i \mid i \geq 0\}$ over the alphabet $\{a, b, c\}$ is :

- (a) a regular language
- (b) not a deterministic context free language but a context free language
- (c) recursive and is a deterministic context free language
- (d) not recursive

Ans : (c) For a language to be recursive it can be decided by Turing machine which means it will enter into final state and reject the rest and for DCFL it must be decidable that for a particular, next character is going to be accepted or not.

25. Suppose we want to download text documents at the rate of 100 pages per second. Assume that a page consists of an average of 24 lines with 80 characters in each line. What is the required bit rate of the channel?

- (a) 192 kbps
- (b) 512 kbps
- (c) 1.248 Mbps
- (d) 1.536 Mbps

Ans : (d) Dynamic data structures are data structures that grow and shrink as you need them to by allocating and deal locating memory from a place called the heap. We have 100 pages, each page is having 24 lines and in each line there are 80 characters and each character is of 8 bits.

Now we have to calculate no. of bits can be downloaded i.e. downloading rate = 100 pages = 100 page * 24 line * 80 character * 8 bits = 1536000 bits/second = 1.536 Mbps

26. Quadrature Amplitude Modulation means changing both :

- (a) Frequency and phase of the carrier
- (b) Frequency and amplitude of the carrier
- (c) Amplitude and phase of the carrier
- (d) Amplitude and wavelength of the carrier

Ans : (c) Quadrature amplitude modulation means changing both amplitude and phase of the Carrier.

27. If a file consisting of 50,000 characters takes 40 seconds to send, then the data rate is

- (a) 1 kbps
- (b) 1.25 kbps
- (c) 2 kbps
- (d) 10 kbps

Ans : (d) 50,000 character takes 40 seconds i.e. $50,000 * 8 = 400,000$ bits (each character is of 8 bits)
In one second $\rightarrow (400,000) / 40$ bits/sec
10,000 bits/sec i.e. 10 kbps

28. Match the following :

List-I		List-II	
A.	Data link layer	i.	Encryption
B.	Network layer	ii.	Connection control
C.	Transport layer	iii.	Routing
D.	Presentation layer	iv.	Framing

Code :

	A	B	C	D
(a)	iv	iii	i	ii
(b)	iii	iv	ii	i
(c)	iv	ii	iii	i
(d)	iv	iii	ii	i

Ans : (d)

- (1) Framing of data in data link layer.
- (2) Network layer is responsible for end to end communication so routing takes place here.
- (3) Connection control is regulated by transport layer.
- (4) Encryption of data takes place at presentation layer.

29. The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and maximum number of hosts in each subnet?

- (a) 62 subnets and 1022 hosts
- (b) 64 subnets and 1024 hosts
- (c) 62 subnets and 254 hosts
- (d) 64 subnets and 256 hosts

Ans : (a) Network Mask of class B = 255:255:0:0

Subnet bits carried from host portion:

$$255.255.\underbrace{111111}_{\text{Subnet}}\underbrace{00:00000000}_{\text{Host}}$$

$$\text{Number of subnet} = 2^6 - 2 = 64 - 2 = 62$$

$$\text{Number of host} = 2^{10} - 2 = 1024 - 2 = 1022$$

30. Which of the following statements are true?

- (a) The fragmentation field in the base header section of IPv4 have moved to the fragmentation extension header in IPv6
- (b) The authentication extension header is new in IPv6
- (c) The record route option is not implemented in IPv6

Code :

- (a) (A) and (B) only
- (b) (B) and (C) only
- (c) (A) and (C) only
- (d) (A), (B) and (C)

Ans : (d) Fragmentation fields in the base header section of IPv4 have moved to the fragmentation extension header in IPv6

0	8	16	29	31
Next header (8 bits)	Reserved (0) (8 bits)	Fragmentation offset (13 bits)	Res (2 bits)	M
Identification (32 bits)				
So all the given statements are correct.				

31. Consider a full binary tree with n internal nodes, internal path length i , and external path length e . The internal path length of a full binary tree is the sum, taken over all nodes of the tree, of the depth of each node. Similarly, the external path length is the sum, taken over all leaves of the tree, of the depth of each leaf. Which of the following is correct for the full binary tree?
- (a) $e = i + n$ (b) $e = i + 2n$
(c) $e = 2i + n$ (d) $e = 2^n + i$

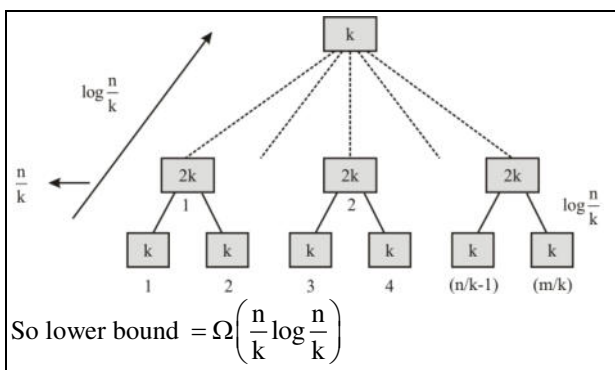
Ans : (b) Consider full binary tree:

Internal nodes = 7

Internal path length = $4 \times 2 + 2 \times 1 = 10$
External path length = $3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 = 24$
So, $e = i + 2n$
 $24 = 10 + 2 \times 7$
 $24 = 10 + 14$
 $24 = 24$

32. You are given a sequence of n elements to sort. The input sequence consists of n/k subsequences, each containing k elements. The elements in a given subsequence are all smaller than the elements in the succeeding subsequence and larger than the elements in the preceding subsequence. Thus, all that is needed to sort the whole sequence of length n is to sort the k elements in each of the n/k subsequences.
- The lower bound on the number of comparisons needed to solve this variant of the sorting problem is :
- (a) $\Omega(n)$ (b) $\Omega\left(\frac{n}{k}\right)$
(c) $\Omega(n \lg k)$ (d) $\Omega\left(\frac{n}{k} \lg \frac{n}{k}\right)$

Ans : (d)



33. Consider the recurrence relation :

$$T(n) = \delta T\left(\frac{n}{2}\right) + Cn, \text{ if } n > 1$$

$$= b, \text{ if } n = 1$$

Where b and c are constants.

The order of the algorithm corresponding to above recurrence relation is :

- (a) n (b) n^2
(c) $n \log n$ (d) n^3

Ans : (d) We can use master theorem to solve this relation $T(n) = a T(n/2) + \theta(n^k \log p n)$, In given question

$$T(n) = 8T(n/2) + cn$$

here $a = 8$ & $b = 2$ and $k = 1$; clearly $a > bk$

$$\text{so } T(n) = \theta(n \log_b a)$$

$$T(n) = \theta(n \log_2 8)$$

$$\text{i.e. } T(n) = o(n^3)$$

34. Consider the following two sequences :

$$X = \langle B, C, D, C, A, B, C \rangle$$

$$\text{and } Y = \langle C, A, D, B, C, B \rangle$$

The length of longest common subsequence of X and Y is :

- (a) 5 (b) 3
(c) 4 (d) 2

Ans : (c)

$$X = \langle B, C, D, C, A, B, C \rangle$$

$$Y = \langle C, A, D, B, C, B \rangle$$

Length of longest common subsequence

$$= \langle C, A, B, C \rangle, \langle C, D, B, C \rangle \text{ etc.}$$

$$= 4$$

35. A text is made up of the characters a, b, c, d, e each occurring with the probability 0.11, 0.40, 0.16, 0.09 and 0.24 respectively. The optimal Huffman coding technique will have the average length of :

- (a) 2.40 (b) 2.16
(c) 2.26 (d) 2.15

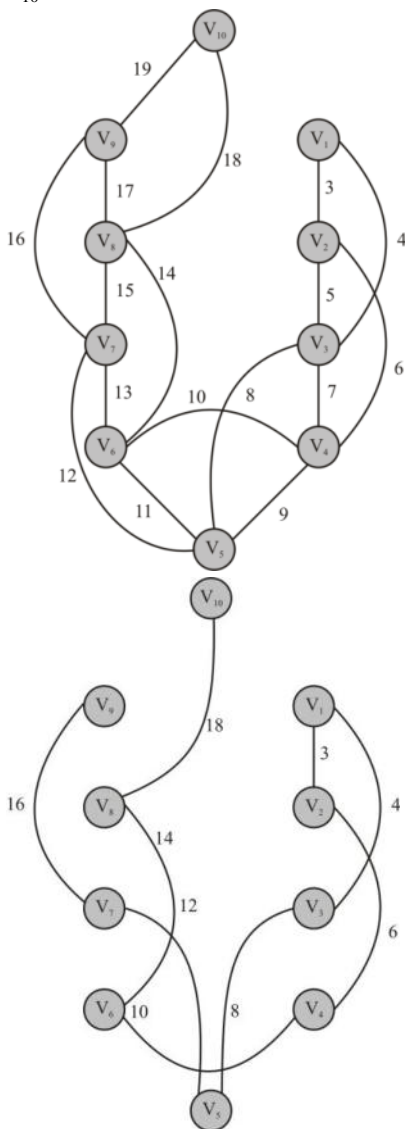
Ans : (b) $a = 0.11, b = 0.40, c = 0.16, d = 0.09, e = 0.24$

we will draw a Huffman tree

Now Huffman coding for character
 $a = 1111$, $b = 0$, $c = 110$, $d = 1111$, $e = 10$;
length for each character = no. of bits * frequency of
occurrence $a = 4 * 0.11 = 0.44$; $b = 1 * 0.4 = 0.4$. $c =$
 $3 * 0.16 = 0.48$ $d = 4 * 0.09 = 0.36$, $e = 2 * 0.24 = 0.48$
Now add these length for average length;
 $0.44 + 0.4 + 0.48 + 0.36 + 0.48 = 2.16$

36. An undirected graph $G(V, E)$ contain n ($n > 2$) nodes named $v_1, v_2, \dots, v_n$. Two nodes v_i and v_j are connected if and only if $0 < |i - j| \leq 2$. each edge (v_i, v_j) is assigned a weight $i + j$
The cost of the minimum spanning tree of such a graph with 10 nodes is :
(a) 88 (b) 91
(c) 49 (d) 21

Ans : (b) Vertices are numbered $V_1, V_2, V_3, V_4, \dots, V_{10}$



Sum = $3 + 4 + 5 + 8 + 10 + 12 + 14 + 16 + 18 = 91$

Note: It is also be calculate by $n^2 - n + 1$.

37. An XML document that adheres to syntax rules specified by XML 1.0 specification in that it must satisfy both physical and logical structured, is called :

- (a) Well - formed (b) Reasonable
(c) Valid (d) Sophisticated

Ans : (a) An XML document that adheres to syntax rules specified by XML 1.0 specification in that it must satisfy both physical & logical structured, is well formed.

38. Which of the following statements (s) is/are TRUE regarding Java serve lets?

(A) A Java servlet is server-side component that runs on the web server and extends the capabilities of a server

(B) A servlet can use the user interface classes like AWT or swing.

Code :

- (a) Only (A) is TRUE
(b) Only (B) is TRUE
(c) Both (A) and (B) are TRUE
(d) Neither (A) nor (B) is TRUE

Ans : (a) A Java servlet is a server side component that runs on the web browser and extends the capabilities of a server. It is correct statement a Java servlet can't be use interfaces classes like AWT or swing.

39. Consider the following HTML table definition :

```
<table border = 1>
  <tr>
    <td colspan = 2>Text A </td>
  </tr>
  <tr>
    <td> Text B </td>
    <td> Text C </td>
  </tr>
  <tr>
    <td rowspan = 2> Text D </td>
    <td> Text E </td>
  </tr>
  <tr>
    <td> Text F </td>
  </tr>
</table>
```

The above HTML code would render on screen as :

- (a)

Text A	
Text B	Text C
Text D	Text E
Text F	
- (b)

Text A	Text B
Text C	
Text D	Text E
	Text F
- (c)

Text A	
Text B	Text C
Text D	Text E
	Text F
- (d)

Text A	
Text B	Text C
Text D	Text E
Text F	

Ans : (c)

Text A

Text B

Text C

Text D

Text E

Text F

40. Which of the following statements is/are TRUE?

- (A) In HTML, character entities are used to incorporate external content into a web page, such as images.
- (B) Once a web server returns a cookie to a browser, the cookie will be included in all future requests from the browser to the same server.

Code :

- (a) Only (A) is TRUE
- (b) Only (B) is TRUE
- (c) Both (A) and (B) are TRUE
- (d) Neither (A) nor (B) is TRUE

Ans : (d) In HTML, character entities are used to incorporate content into a web page such as images. So this statement is wrong.

Once a web server return a cookie to a browser, the cookie will be included in all future requests from the browser to the same server. So this is also a wrong one.

41. Which of the following statements is/are TRUE regarding JAVA?

- (A) Constants that cannot be changed are declared using the 'static' keyword
- (B) A class can only inherit one class but can implement multiple interfaces

Code :

- (a) Only (A) is TRUE
- (b) Only (B) is TRUE
- (c) Both (A) and (B) are TRUE
- (d) Neither (A) nor (B) is TRUE

Ans : (b) In java, content are not declared using 'static' keyword and a class can implement multiple interfaces but class can inherit one class only.

42. What is the output of the following JAVA program?

Class Test

{

public static void main (String [] args)

{

Test obj = new Test ();

obj. start () ;

}

void start ()

{

String stra = "do";

String strb = method (stra);

System.out.print(":" + stra + strb);

}

String method (string stra)

```

{
    stra = stra + "good";
    System.out.print(stra);
    return "good";
}
}

```

(a) dogood : dogoodgood
(b) dogood : gooddogood
(c) dogood : dodogood
(d) dogood : dogood

Ans : (d) The output of the given Java program is dogood : dogood .

43. Statistical software quality assurance in software engineering involves.....

- (a) using sampling in place of exhaustive testing of software
(b) surveying customers to find out their opinions about product quality
(c) tracing each defect to its underlying cause, isolating the vital few causes, and moving to correct them
(d) tracing each defect to its underlying causes, and using the pare to principle to correct each problem found

Ans : (c) Statistical software quality assurance in software engineering involves tracing each defect to its underlying cause, isolating the vital few causes, and moving to correct them.

44. Which of the following statements is/are FALSE with respect to software testing?

S1 : White-box tests are based on specifications; better at telling whether program meets specification, better at finding errors of omission

S2 : Black-box tests are based on code; better for finding crashes, out of bounds errors, file not closed errors.

S3 : Alpha testing is conducted at the developer's site by a team of highly skilled testers for software that is developed as a product to be used by many customers

- (a) Only S1 and S2 are FALSE
(b) Only S1 and S3 are FALSE
(c) Only S2 and S3 are FALSE
(d) All of S1, S2 and S3 are FALSE

Ans : (a) Alpha testing is conducted at the developer's site by a team of highly skilled testers for software that is developed as a product to be used by many customers statement about white box testing and black box testing are not correct. Only the last statement is correct.

45. A signal processor software is expected to operate for 91.25 days after repair, and the mean software repair time is expected to be 5 minutes. Then, the availability of the software is :

- (a) 96.9862% (b) 97.9862%
(c) 98.9962% (d) 99.9962%

Ans : (d)

$$\text{Availability} = \frac{\text{MTBF}}{(\text{MTBF} + \text{MTRR})}$$

$$= \left(\frac{91.25 \times 24 \times 60}{91.25 \times 24 \times 60 + 5} \right)$$

$$\text{Availability} = 99.9962\%$$

46. Consider the method mcq () :

```
int mcq (boolean a, boolean b, boolean c,
boolean d)
{
    int ans = 1;
    if (a) {ans = 2;}
    else if (b) {ans = 3;}
    else if (c) {
        if (d) {ans = 4;}
    }
    return ans;
}
```

M1 = Number of tests to exhaustively test mcq ();

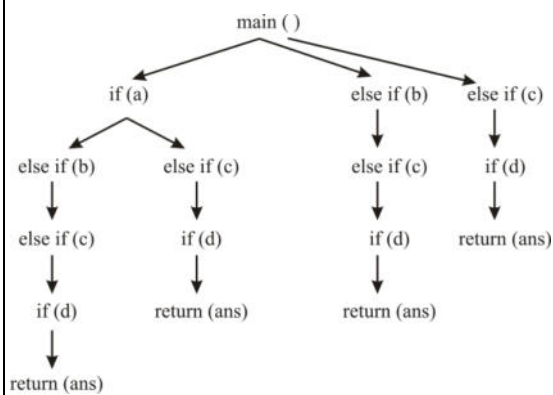
M2 = Minimum number of tests to achieve full statement coverage for mcq (); and

M3 = Minimum number of tests to achieve full branch coverage for mcq ();

then (M1, M2, M3) =

- (a) (16, 3, 5) (b) (8, 5, 3)
 (c) (8, 3, 5) (d) (16, 4, 4)

Ans : (d)



So, number of test to exhaustively test = 16

Full statement coverage = 4

Full branch coverage for = 4

47. A simple stand – alone software utility is to be developed in 'C' programming by a team of software experts for a computer running Linux and the overall size of this software is estimated to be 20, 000 lines of code. Considering (a, b) = (2.4, 1.05) as multiplicative and exponention factor for the basic COCOMO effort estimation equation and (c, d) = (2.5, 0.38) as multiplicative and exponention factor for the basic COCOMO development time estimation

equation, approximately how long does the software project take to complete?

- (a) 10.52 months (b) 11.52 months
(c) 12.52 months (d) 14.52 months

Ans : (b) Total lines codes is 20,000 and (a, b) = (2.4k, 1.05) and (c, d) = (2.5, 0.38) are the multiplicative & exoneration factor for basis COCOMO development time equation. After that the total software take to complete is 11.52 months.

48. In software configuration Management (SCM), which of the following is a use-case supported by standard version control system?

- (a) Managing several versions or release of a software
(b) Filing bug reports and tracking their progress
(c) allowing team members to work in parallel
(d) Identifying when and where a regression occurred

Code :

- (a) Only (A), (C), and (D)
(b) Only (A), (B) and (C)
(c) Only (A), (B) and (D)
(d) Only (B), (C) and (D)

Ans : (a) Managing several version of software is managed by SCM filling bug reports & tracking their progresses not managed by SCM. Allowing team members to work in parallel is managed by SCM. Identifying when and where a regression occurred is managed by SCM.

49. Consider the following four processes with the arrival time and length of CPU burst given in milliseconds :

Process	Arrival Time	Burst Time
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

The average waiting time for preemptive SJF scheduling algorithm is

- (a) 6.5 ms (b) 7.5 ms
(c) 6.75 ms (d) 7.75 ms

Ans : (a) First we will make Gantt chart of given process then we will calculate turn around time and waiting time of individual process.

I					
P1	P2	P4	P1	P3	
0	1	5	10	17	26

Process id	CT	TAT	WT
P1	17	16	9
P2	5	4	0
P3	26	24	15
P4	10	7	2

Now we have calculate average waiting time for
 schedule avg waiting time = $W_T(P1 + P2 + P3 + P4)/4$
 $= 26/4 = 6.5$

- 50. Consider a virtual page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. Suppose a demand paged virtual memory system running on a computer system such that the main memory has 3 page frames. Then..... page replacement algorithm has minimum number of page faults.**
- (a) FIFO (b) LIFO
 (c) LRU (d) Optimal

Ans : (d) In optimal algorithm 9 page fault will occur

7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
		1	1	1	3	3	3	3	3	3	3	3	1	1	1	1	1	1	1
7	0	0	0	0	0	0	4	4	4	0	0	0	0	0	0	0	0	0	0
7	7	7	2	2	2	2	2	2	2	2	2	2	2	2	2	2	7	7	7
F	F	F	F		F		F		F		F		F		F				

and if FIFO algorithm is fault will occur & in LIFO 11 page fault will occur & in LRU 12 page fault will occur. So the minimum page fault will occur in optimal algorithm.

- 51. User level threads are threads that are visible to the programmer and are unknown to the kernel. The operating system kernel supports and manages kernel level threads. Three different types of models relate user and kernel level threads.**
- Which of the following statements is/are true?**
- (A) (i) The many to one model maps many user threads to one kernel thread
 (ii) The one - to - one model maps one user threads to one kernel thread
 (iii) The many model maps many user threads to smaller or equal kernel threads
- (B) (i) Many - to- one model maps many kernel threads to one user thread
 (ii) One - to - one model maps one kernel thread to one user thread
 (iii) Many - to - many model maps many kernel threads to smaller or equal user threads

Code :

- (a) (A) is true; (B) is false
 (b) (A) is false; (B) is true
 (c) Both (A) and (B) are true
 (d) Both (A) and (B) are false

Ans : (a)

(a) (i) The may to one model maps many user thread to one kernel thread.
 (ii) The one to one model maps one user thread to one kernel thread.
 (iii) The many to many model maps many user thread to smaller or equal kernel threads.

(b) (i) Many to one model maps many kernel thread to one user thread.

(ii) One to one model maps one kernel thread to one user thread.

(iii) Many to many model maps many kernel thread to smaller or equal user thread

So all statements are correct hence option (a) is answer.

52. Consider a system with five processes P₀ through P₄ and three resource types A, B and C. Resource type A has seven instances, resource type B has two instances and resource type C has six instances. Suppose at time T₀ we have the following allocation.

Process	Allocation			Request			Available		
	A	B	C	A	B	C	A	B	C
P ₀	0	1	0	0	0	0	0	0	0
P ₁	2	0	0	2	0	2			
P ₂	3	0	3	0	0	0			
P ₃	2	1	1	1	0	0			
P ₄	0	2	2	0	0	2			

If we implement Deadlock detection algorithm we claim that system is

- (a) Semaphore (b) Deadlock state
(c) Circular wait (d) Not in deadlock state

Ans : (d)

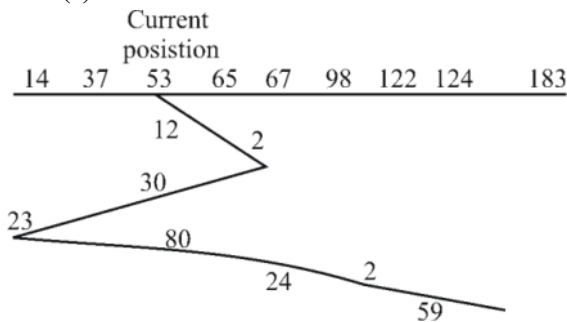
Process	Allocation			Request			Available		
	A	B	C	A	B	C	A	B	C
							0	0	0
P ₀	0	1	0	0	0	0	3	1	3
P ₁	2	0	0	2	0	2	5	1	3
P ₂	3	0	3	3	0	3	3	0	3
P ₃	2	1	1	2	1	1	7	2	4
P ₄	0	2	2	0	0	2	7	4	6

So, not in deadlock state.

53. Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 37, 122, 14, 124, 65, 67. Suppose SSTF disk scheduling algorithm implemented to meet the requests then the total number of head movements are if the disk head is initially at 53.

- (a) 224 (b) 248
(c) 236 (d) 240

Ans : (c)



Now add all seek time

$$12 + 2 + 30 + 23 + 84 + 24 + 2 + 59 = 236$$

- 54. The Bounded buffer problem is also known as.....**
- (a) Producer - consumer problem
 - (b) Reader - writer problem
 - (c) Dining philosophers problem
 - (d) Both (2) and (3)

Ans : (a) The producer consumer problem (or bounded buffer problem) describes two processes, the producer and the consumers which share a common, fixed size buffer used as a queue. Producer produces an item & put into buffer. If buffer is already full then producer will wait for an empty block in buffer consumer consumes an item from buffer.

- 55. In Artificial Intelligence (AI), which agent deals with happy and unhappy state?**
- (a) Simple reflex agent
 - (b) Model based agent
 - (c) Learning agent
 - (d) Utility based agent

Ans : (b) Model based agent deals with happy & unhappy state in artificial intelligence (AI). So option (b) is correct

- 56. If b is the branching factor and m is the maximum depth of the search tree, what is the space complexity of greedy search?**
- (a) $O(b + m)$
 - (b) $O(bm)$
 - (c) $O(b^m)$
 - (d) $O(m^b)$

Ans : (c) (1) In binary tree branching factor is 2 and space complexity of height n is $O(2^n)$
 (2) In ternary tree branching factors is 3 & space complexity for height n is $O(3^n)$ similarly
 (3) If branching factor is b and height is m for search tree then space complexity for greedy search is $O(b^n)$.

- 57. Let P, Q, R and S be propositions. Assume that the equivalences $P \Leftrightarrow (Q \vee \neg Q)$ and $Q \Leftrightarrow R$ hold. Then the truth value of the formula $(P \wedge Q) \Rightarrow (P \wedge R) \vee S$ is always :**
- (a) True
 - (b) False
 - (c) Same as truth table of Q
 - (d) Same as truth of S

Ans : (a)

1. $P \Leftrightarrow (Q \vee \neg Q) \equiv P \Leftrightarrow 1 = P'$
 2. $Q \Leftrightarrow R \equiv Q'R' + QR$
 Given $(P \wedge Q) \Rightarrow ((P \wedge R) \vee S)$
 $(P \wedge R) \Rightarrow ((P \wedge R) \vee S)$
 $(PR)' + PR + S$
 $= P' + R' + PR + S$
 $= P' + R + R' + S = 1$
 So, always true.

58. "If X, then Y unless Z" is represented by which of the following formulae in propositional logic?

- (a) $(X \wedge Y) \rightarrow \neg Z$
- (b) $(X \wedge \neg Z) \rightarrow Y$
- (c) $X \rightarrow (Y \wedge \neg Z)$
- (d) $Y \rightarrow (X \wedge \neg Z)$

Ans: (b) If X then Y is represented by $X \rightarrow Y$ one more condition is unless Z, it will be included in X, so X then Y unless Z is represented by $(X \wedge \neg Z) \rightarrow Y$

59. Consider the following two well-formed formulas in propositional logic.

F1 : $P \Rightarrow \neg P$

F2 : $(P \Rightarrow \neg P) \vee (\neg P \Rightarrow P)$

Which of the following statements is correct?

- (a) F1 is satisfiable, F2 is valid
- (b) F1 is unsatisfiable, F2 is satisfiable
- (c) F1 is unsatisfiable, F2 is valid
- (d) F1 and F2 both are satisfiable

Ans : (b) $F1 : P \rightarrow \neg P$
 $= \neg P(\neg P) + P \neg (\neg P)$
 $= \neg P + P$ is satisfiable
 $F2 : (P \Rightarrow \neg P) \vee (\neg P \Rightarrow P)$
 $\Rightarrow (\neg P \neg P) + P \neg (\neg P) \vee (P \neg (\neg P) + \neg P (\neg P))$
 $= \text{satisfiable} \vee \text{satisfiable} = \text{satisfiable}$
 So option (a) is correct

60. Standard planning algorithms assume environment to be.....

- (a) Both deterministic and fully observable
- (b) Neither deterministic nor fully observable
- (c) Deterministic but not fully observable
- (d) Not deterministic but fully observable

Ans : (a) Standard planning algorithms assume environment to be both deterministic and fully observable.

61. Which of the following statements is not correct?

- (a) Every recursive language is recursively enumerable
- (b) $L = \{0^n 1^n 0^n \mid n = 1, 2, 3, \dots\}$ is recursively enumerable
- (c) Recursive language are closed under intersection
- (d) Recursive language are not closed under intersection.

Ans : (d) Recursive language are closed under intersection so option (d) is correct.

62. Context free grammar is not closed under :

- (a) Concatenation
- (b) Complementation
- (c) Kleene Star
- (d) Union

Ans : (b) Context free grammar are not closed under intersection context free language are accepted by pushdown automata but not by finite automata - context free grammar.

63. Consider the following language :

$$L_1 = \{a^m b^n \mid m \neq n\}$$

$$L_2 = \{a^m b^n \mid m = 2n + 1\}$$

$$L_3 = \{a^m b^n \mid m \neq 2n\}$$

Which one the following statement is correct?

- (a) Only L_1 and L_2 are context free languages
- (b) Only L_1 and L_3 are context free languages
- (c) Only L_2 and L_3 are context free languages
- (d) L_1 , L_2 and L_3 are context free languages

Ans : (d) (1) In L_1 we have to check for m is not equal to n there are infinite number which falls under this category but we have only one comparison with m and n . So it is context free language

(2) In L_2 one comparison with infinite possibilities of m and n it also a CFL

(3) It is also a CFL

64. A 4×4 DFT matrix is given by :

$$\frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & x & -1 & y \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

$$(j^2 = -1)$$

Where values of x and y are, respectively.

- (a) $1, -1$
- (b) $-1, 1$
- (c) $-j, j$
- (d) $j, -j$

Ans : (d) A 4×4 discrete fourier transformation will be look like:

$$\frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & x & -1 & y \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \quad [\text{where } j^2 = -1]$$

$$\begin{aligned} \text{Here,} & \quad w = x \text{ and } w^3 = y \\ \text{i.e.} & \quad w^2 = -1 \text{ given on comparing} \\ \text{So} & \quad w^2 = j^2 \\ \text{And} & \quad w = j \text{ i.e. } x = j \\ & \quad w^3 = w \cdot w^2 \\ & \quad = x - 1 = j, -j \end{aligned}$$

65. Entropy of a discrete random variable with possible values $\{x_1, x_2, \dots, x_n\}$ and probability density function $P(X)$ is :

$$H(X) = -\sum_{i=1}^n P(x_i) \log_b P(x_i)$$

The value of b gives the units of entropy. The unit for $b = 10$ is :

- (a) bits
- (b) bann
- (c) nats
- (d) deca

Ans : (c) Information entropy or Shannon's entropy quantities the amount of uncertainty involved in the value of a random variable or the outcome of a random process.

66. For any binary (n, h) linear code with minimum distance $(2t + 1)$ or greater $n - h \geq$

$$\log_2 \left[\sum_{i=0}^{\alpha} \binom{n}{i} \right] \text{ where } \alpha \text{ is :}$$

- (a) $2t + 1$
- (b) $t + 1$
- (c) $t - 1$
- (d) t

Ans : (b) Answer is $t + 1$ means .

67. Which of the following is a valid reason for causing degeneracy in a transportation problem? Here m is no. of rows and n is no. of columns in transportation table

- (a) When the number of allocation is $m + n - 1$
- (b) When two or more occupied cells become unoccupied simultaneously
- (c) When the number of allocations is less than $m + n - 1$
- (d) When a loop cannot be drawn without using unoccupied cells, except the starting cell of the loop

Ans : (c) When the number of allocations is less than $m + n - 1$, It causes degeneracy in transpiration problem.

68. Consider the following LPP :

$$\text{Max } Z = 15x_1 + 10x_2$$

Subject to the constraints

$$4x_1 + 6x_2 \leq 360$$

$$3x_1 + 0x_2 \leq 180$$

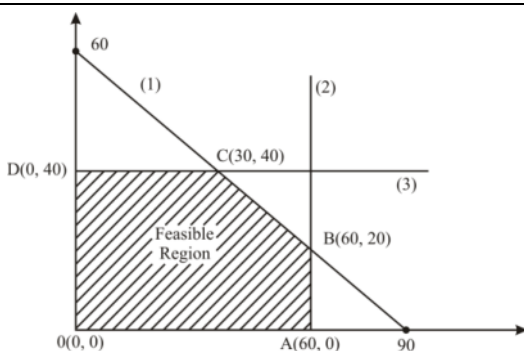
$$0x_1 + 5x_2 \leq 200$$

$$x_1, x_2 \geq 0$$

The solution of the LPP using Graphical solution technique is :

- (a) $x_1 = 60, x_2 = 0$ and $Z = 900$
- (b) $x_1 = 60, x_2 = 20$ and $Z = 1100$
- (c) $x_1 = 60, x_2 = 30$ and $Z = 1200$
- (d) $x_1 = 50, x_2 = 40$ and $Z = 1150$

Ans : (b)



$$4x_1 + 6x_2 = 360$$

$$3x_1 = 180$$

$$5x_2 = 200$$

Z is maximum at a vertex of a feasible region, which is polygon, given in table at O, A, B, C and D.

Extreme Point	Co-ordinates	Z
O	(0, 0)	0
A	(60, 0)	900
B	(60, 20)	1110
C	(30, 40)	850
D	(0, 40)	400

Maximum Z = 1110 at the point (60, 20).

69. Consider the following LPP :

$$\text{Min } Z = 2x_1 + x_2 + 3x_3$$

Subject to :

$$x_1 - 2x_2 + x_3 \geq 4$$

$$2x_1 + x_2 + x_3 \leq 8$$

$$x_1 - x_3 \geq 0$$

$$x_1 - x_2, x_3 \geq 0$$

The solution of this LPP using Dual Simplex method is :

(a) $x_1 = 0, x_2 = 0, x_3 = 3$ and $Z = 9$

(b) $x_1 = 0, x_2 = 6, x_3 = 0$ and $Z = 6$

(c) $x_1 = 4, x_2 = 0, x_3 = 0$ and $Z = 8$

(d) $x_1 = 2, x_2 = 0, x_3 = 2$ and $Z = 10$

Ans : (c) Ist step in dual simplex method is to convert the given problem into canonical form.

In canonical form given function should be of max type

Thus, to make it max, multiply both sides by -1 , This objective becomes $\max (-Z) = -2x_1 - x_2 - 3x_3$

subject to

$$-x_1 + 2x_2 - x_3 \leq -4$$

$$2x_1 + x_2 + x_3 \leq 8$$

$$-x_1 + x_3 \leq 0$$

& the second step is to write standard form & then iterate it and in the last the value is $z = 8, x_1 = 4, x_2 = 0, x_3 = 0$.

70. Consider a Takagi – Sugeno – Kang (TSK)

Model consisting of rules of the form :

If x_1 is A_{i1} and..... and x_r is A_{ir}

THEN $y = f_i(x_1, x_2, \dots, x_r) = b_{i0} + b_{i1} x_1 + \dots + b_{ir} x_r$

assume, α_i is the matching degree of rule i , then the total output of the model is given by :

$$(a) \quad y = \sum_{i=1}^L \alpha_i f_i(x_1, x_2, \dots, x_r)$$

$$(b) \quad y = \frac{\sum_{i=1}^L \alpha_i f_i(x_1, x_2, \dots, x_r)}{\sum_{i=1}^L \alpha_i}$$

$$(c) \quad y = \frac{\sum_{i=1}^L f_i(x_1, x_2, \dots, x_r)}{\sum_{i=1}^L \alpha_i}$$

$$(d) \quad y = \max_i [\alpha_i f_i(x_1, x_2, \dots, x_r)]$$

Ans : (b)

$$y = \frac{\sum_{i=1}^L \alpha_i f_i(x_1, x_2, \dots, x_r)}{\sum_{i=1}^L \alpha_i}$$

71. Consider a single perceptron with sign activation function. The perceptron is represented by weight vector $[0.4 \ -0.3 \ 0.1]^T$ and a bias $\theta = 0$. If the input vector to the perceptron is $X = [0.2 \ 0.6 \ 0.5]$ then the output of the perceptron is :

- (a) 1 (b) 0
(c) -0.05 (d) -1

Ans : (c) Here it is given input vector = $[0.2 \ 0.6 \ 0.5]$ & weight vector $w = [0.4 \ -0.3 \ 0.1]^T$

The output of the perception will be the summation of x & w represented as,

$$\Sigma : X_i W_i \quad (i = 1, 2, 3 \text{ because there are 3 values of } X \text{ \& } W) \\ [0.2 \ 0.6 \ 0.5] * [0.4 \ -0.3 \ 0.1] = [0.2 * 0.4 \ 0.6 * (-0.3) \ 0.5 * 0.1] = \\ [0.08 - 0.18 + 0.05] = 0.05$$

72. The Sigmoid activation function $f(t)$ is defined as :

- (a) $\frac{1}{\exp(t) + \exp(-t)}$ (b) $t \exp(-t)$
(c) $\frac{1}{1 + \exp(t)}$ (d) $\frac{1}{1 + \exp(-t)}$

Ans : (c) The sigmoid activation function $f(t)$ is defined as $1/(1 + \exp(t))$.

- (A) UNIX provides three types of permissions**
 - * Read**
 - * Write**
 - * Execute**
- (B) UNIX provides three sets of permissions**
 - * permission for owner**
 - * permission for group**
 - * permission for others**

(a) Only (A)
(b) Only (B)
(c) Both (A) and (B)
(d) Neither (A) nor (B)

**74. Which of the following routing technique/
techniques is/are used in distributed systems?**

(A) Fixed Routing
(B) Virtual Routing
(C) Dynamic Routing

Code :

(a) (A) only (b) (A) and (B) only
(c) (C) only (d) All (A), (B), (C)

75. Match the following WINDOWS system calls and UNIX system calls with reference to process control and file manipulation.

Windows		UNIX	
A.	Create - process ()	i.	Open ()
B.	Wait for single object ()	ii.	Close ()
C.	Create file ()	iii.	Fork ()
D.	Close Handle ()	iv.	Wait ()

Ans : (b) 1. In UNIX wait () is used to create process ()
2. In UNIX fork () is used for wait for single object ()
3. In UNIX create () is used for create file ()
4. In UNIX close () is used for close handle ()